

Project Outline — FitPilot (AI Fitness Coach)

1. APP NAME

FitPilot — Your AI Fitness Coach that *sees*, *thinks* and *tracks*.

2. App Category

Health & Fitness

3. About the App

FitPilot is an Android application that acts as a **personal AI fitness coach**. Unlike traditional fitness apps where the user has to pick a course and follow it, FitPilot **decides what you should train today** and **watches you while you do it**.

The app combines three capabilities into one closed loop:

1. **It thinks (Daily Decision Engine)**: Every morning, FitPilot recommends "what to train today" by combining the **weather & air quality** (e.g. rainy or polluted AQI → indoor/bodyweight session), the user's **schedule & preferred time**, their **Recovery Score** (from resting heart rate, sleep and past workouts), and their **training preference** (gym equipment vs. home bodyweight). A short onboarding questionnaire captures the user's **training scene (home vs gym) and equipment inventory** (dumbbells? treadmill? bodyweight-only?) so the engine can hard-filter the exercise library to what the user can actually do.
2. **It sees (Real-time AI Form Coach)**: During strength training (e.g. bench press), the app uses the phone camera with an on-device pose-estimation model (TensorFlow Lite + MoveNet) to **count reps automatically and judge whether each rep is standard**, giving instant voice feedback (e.g. "Rep 5 — keep your elbows tucked").
3. **It tracks (Unified Training Log)**: During running it records **heart rate** (BLE chest strap) and **distance** (GPS); during strength training it records **reps, sets and form quality**; it also logs **jump-rope counts** and **HIIT intervals**. All data is stored locally and feeds back into smarter future recommendations.

Why there is a gap (Niche): Existing apps cover parts of this idea but not the combination — Fitbod generates daily plans based on muscle recovery but ignores weather and gives no real-time form feedback; Keep has AI form scoring and follow-along courses but keeps running and strength as two separate systems, and its plans are "pick & play" rather than "decided for you"; Good Habits (open source) proves on-device rep counting works but does no recommendation at all. FitPilot is the first **fully local, explainable** coach that unifies **weather/AQI-aware daily decisions + a recovery score + real-time multi-exercise form feedback + unified multi-scenario training logs**.

4. Existing Similar Apps with Source Code

App	App Details	Source Code
Fitbod	Generates personalized daily workout plans based on muscle recovery, equipment availability and progress history. Cloud subscription, no weather dimension, no live form feedback.	Not open source
Keep	Largest Chinese fitness platform: smart training plans, AI form scoring on follow-along videos, running + strength content — but the two scenarios are separate systems , plans are course-selection based, and AI scoring is server-side.	Not open source
Good Habits <i>(chosen base)</i>	100% native Android (Kotlin + Jetpack Compose + Room + MVVM), on-device TensorFlow Lite pose estimation with 90+ exercises, rep counting, streak tracking, CSV export. Rep counting only — no recommendation, no weather, no voice coaching.	github.com/plana93/good-habits
HeartRunner	Running app with BLE heart-rate strap, GPS track, pace stats and Chinese voice feedback; clean Compose + MVVM architecture. Running only — no strength tracking, no recommendation.	github.com/igofreely/runner (MIT)
wger	Open fitness ecosystem: exercise database (500+), workout manager, free REST API. Web/Flutter based, not native Android; no AI.	github.com/wger-project/wger (AGPL)

App chosen to build upon: Good Habits (primary — pose-AI architecture, data layer and UI patterns are directly reusable), complemented by patterns from **HeartRunner** (BLE heart rate + GPS + TTS voice feedback). The exercise dataset is bundled locally (from Good Habits' built-in library and/or open exercise JSON data) so the app works **fully offline**.

5. Main New Added Features & Enhancements

Compared with the chosen base apps, FitPilot adds:

1. **User profile & equipment-inventory-driven recommendation engine** — a short onboarding questionnaire captures **training scene (home vs gym), equipment inventory (dumbbells / treadmill / bodyweight-only), training goal (fat loss / muscle gain / toning), weekly frequency and experience level**. Each exercise in the library carries tags (type: cardio vs strength; equipment; muscle group; difficulty), and the engine **hard-filters**

out exercises the user cannot do, then ranks the rest. It also pulls **weather + air-quality (QWeather API)** so a polluted AQI automatically switches an outdoor run to an indoor plan.

2. **Recovery Score (0–100)** — a transparent score combining morning resting heart rate (BLE or manual), sleep & subjective fatigue check-in, and recent training load. It drives daily intensity and triggers overtraining warnings (e.g. "recovery low for 3 days → rest day").
3. **Real-time form-quality checking** — beyond counting reps, computes joint angles (shoulder/elbow/knee) to judge whether a rep is standard and gives corrective feedback; applied to bench press, squat and plank.
4. **Jump-rope auto counting** — pose-based detection of jump cycles for a cardio session, with on-screen count and voice milestones.
5. **Voice coaching feedback** — Android built-in TTS announces rep counts, correction hints, HIIT interval commands ("30s sprint / 15s rest") and a guided stretching cool-down (no cloud needed).
6. **Unified multi-scenario closed loop** — running (HR + distance), strength (reps + form), jump rope and HIIT data are unified in one Room database and feed future recommendations (recovery-aware intensity adjustment).
7. **Fully local & explainable AI** — pose estimation and the recommendation algorithm both run on-device, free, offline, and every decision is traceable.
8. *(Planned enhancement)* **LLM-assisted weekly summary** — a Chinese LLM API (e.g. DeepSeek) summarizes the week's training into natural language; non-critical path.

6. Functional Requirements (App Features)

Ser	Existing Features (from chosen app)	Proposed Improvements / New Features
1	Exercise library + manual workout logging (Good Habits)	NEW: Onboarding questionnaire (scene / equipment / goal / level) → tagged exercise library + hard-filter by equipment
2	Rep counting via on-device pose AI (Good Habits)	IMPROVED: Form-quality judgement (joint angles) for bench press / squat / plank + jump-rope auto counting
3	—	NEW: Running session: GPS distance, pace, BLE heart-rate recording with live chart
4	—	NEW: Recovery Score (resting HR + sleep + fatigue check-in) with overtraining warning
5	—	NEW: HIIT interval voice coach + guided stretching cool-down
6	—	NEW: Closed loop — past records automatically adjust next day's recommended intensity
7	Streak / statistics (Good Habits)	IMPROVED: Unified charts & recovery trend visualization across all scenarios

7. Tool Stack

Purpose	Tool / API	Notes
Weather & air data	QWeather (和风天气)	Free monthly quota, direct access in mainland China, no VPN , provides weather icons, air-quality index (AQI) and sport/comfort indices used by the recommendation engine
Pose estimation	TensorFlow Lite + MoveNet	On-device ML, offline, free — the same approach proven by Good Habits
Local database	Room (SQLite)	Exercises (with tags), workouts, running sessions, user profile & equipment inventory, recovery state
UI	Jetpack Compose + Material 3	Modern declarative UI, dark mode
Network	Retrofit + OkHttp	QWeather API calls
Location	高德地图 SDK / osmdroid	GPS track for running (mainland-accessible)
Heart rate	Android BLE API	Connect to standard BLE heart-rate chest strap
Voice feedback	Android TextToSpeech	Built-in, offline, Chinese supported
Architecture	MVVM + Repository	Follows Good Habits' clean structure
Version control	Git + GitHub	Team collaboration & commit history
<i>(Planned)</i> LLM	DeepSeek / 通义千问	Optional weekly summary only — non-critical path

8. Development Timeline

Week	Work to be Completed	Comments
6	Scaffold Android project on the Good Habits base (Compose + Room + MVVM); design DB schema; import tagged exercise dataset (type / equipment / muscle / difficulty); onboarding questionnaire (scene / equipment / goal / frequency / level)	Sprint to Alpha
7	Alpha submission — recommendation engine v1 (equipment hard-filter + weighted scoring) + basic workout logging working	Milestone

8	MoveNet pose integration: bench-press rep counting + joint-angle form judgement	Sprint to Beta
9	TTS voice feedback; running session: GPS distance & pace (osmdroid/高德) + BLE heart-rate strap; live HR chart	
10	Beta submission — real-time form feedback + running/heart-rate loop complete	Milestone
11	Recovery Score (sleep / fatigue check-in + resting HR) → closed-loop intensity adjustment	
12	QWeather AQI integration; unified history & recovery-trend charts; UI polish / dark mode	
13	Edge cases (no network / no BLE / no camera permission); documentation; demo script; <i>(if time)</i> jump-rope / HIIT / stretching	Optional extras
14	Final submission — video + report + live demo	Milestone

9. Key Contributions (Summary)

1. A **weather + AQI-aware daily decision engine** that existing fitness apps lack (polluted AQI → indoor plan).
2. A **Recovery Score** that turns raw logs into a transparent "how fresh am I" number, driving daily intensity and rest-day warnings.
3. **Real-time form feedback with voice coaching** across strength (bench press / squat / plank), jump rope and HIIT — fully on-device and explainable.
4. A **unified multi-scenario training log** (running + strength + HIIT) that closes the data loop into smarter recommendations.